



A model for vortex-induced vibration analysis of long-span bridges

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ABSTRACT

Long-span structures are susceptible to wind-induced vibrations due to their low oscillation frequency and low mechanical damping. Although many efforts have been made in the past to model vortex-induced vibration of circular cylinders, limited studies can be found for non-circular cross sections representative of long-span bridge decks. A model for vortex-induced vibration analysis of long-span bridge is presented in this paper. The aeroelastic equation of motion of the model, a procedure to extract aeroelastic coefficients from wind tunnel experiments, analysis of full-scale structures incorporating loss of spanwise correlation of aeroelastic forces, and comparison between simulated and full-scale measured responses on a twin deck bridge (Fred Hartman bridge, Baytown, Texas) are discussed. Six bluff sections – Deer Isle bridge, Tsurumi bridge, Fred Hartman bridge, generic rectangular, H shaped, and circular models – were considered in this research program.

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1. Introduction

The study of flow-induced vibration of long-span civil engineering structures has emerged as an important field after the collapse of the Tacoma Narrows bridge in 1940 (Billah and Scanlan, 1991). Since then significant progress has been made in the understanding of aerodynamic and aeroelastic phenomena specially flutter, buffeting, and galloping vibration of long-span civil structures and various analytical tools for analyzing structural responses have been developed, which are now routinely implemented in design. However, despite some efforts made in the past, vortex-induced vibration of non-circular long-span structures still challenges structural and wind engineers. When flow is separated by a bluff body, a regular pattern of vortices develops in the downstream region that exerts periodic pressure on the body. The frequency (f_s) of vortex shedding may approach one of the natural frequencies (f_n) of a flexible structure at some critical wind speeds. If the ratio f_s/f_n becomes close to unity, the structure and the wake begin to oscillate at a common frequency close to the natural frequency of the structure over a certain band of wind speeds. The oscillation frequency of the structure takes control of the shedding frequency over the bandwidth; as a result, shedding frequency violates the linear Strouhal relationship. The overall phenomenon described above is sometimes called “locked-in” vibration and the region within which it occurs is often called the “lock-in” region.

This type of vibration can generate moderately large amplitude response on a structure.

Locked-in vibration was first carefully investigated by Bishop and Hassan (1964) by subjecting a circular cylinder to forced oscillation in water and then by Feng (1968) on a spring-mounted rigid circular and D-shape sections in air. Motivated by the interesting findings of these pioneering experimental investigations, Hartlen and Currie (1970) first proposed a mathematical model similar to the Van der Pol equation, which was originally developed to idealize a spontaneously oscillating but self-limiting valve circuit (Van der Pol, 1926). Thereafter, a number of researchers had developed different models for predicting locked-in vibration of circular cylinders (Skop and Griffin, 1975; Iwan and Blevins, 1974; Landl, 1975; Sarpkaya, 1978; Dowell, 1981; Vickery and Basu, 1983; Staubli, 1983; Billah, 1989; Goswami et al., 1993). As the models for circular cylinders cannot be used directly for the analysis of non-circular sections due to considerable differences between the characteristics of locked-in vibrations of circular and non-circular sections, a few researchers proposed mathematical models suitable for long-span bridges (Scanlan, 1981; Simiu and Scanlan, 1986; Ehsan and Scanlan, 1990). One of the major limitations of the existing models for full-scale bridges is that these models are not rigorously developed like those for flutter and buffeting. Although Ehsan (1989) simulated the responses of several long-span bridges using the Scanlan–Simiu model, the results were not validated with full-scale measurements. Additionally, the Scanlan–Simiu model employs a Van der Pol type equation, as was done in the Hartlen–Currie and other models, without much physical justification. A few recent events of vortex-induced vibration on prototype bridges and a general consensus found in technical articles and reports (Fujino and Yoshida, 2002; Larsen, 1993; Irwin, 1998;

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Larsen et al., 2000) on the lack of good analytical models have led to the conclusion that a model capable of reliably predicting the response of a full-scale bridge is much needed. Additionally, research on locked-in vibration of twin-deck bridge sections is limited.

A systematic approach was considered towards developing the model. Wind tunnel experiments were first performed on several spring-mounted rigid models, including a twin-deck bridge model, to advance the understanding of the mechanism of vortex-induced vibration of non-circular sections and collect necessary data. Based on the experimental observations, a practical mathematical model was developed and associated experimental procedure was established. A modal analysis technique was developed for predicting the locked-in response of prototype structures, which was used to predict the response of the Fred Hartman bridge. Simulated locked-in response of the bridge was validated by long-term, full-scale data measured on the bridge.

2. Summary of experiments

Experiments were performed in an open-return, low-speed, low-turbulence wind tunnel in the Subsonic Aerodynamics Laboratory at the University of Illinois, Urbana-Champaign. The working cross section of the tunnel is 0.9 m × 1.3 m and maximum attainable speed is 71 m/sec. A force-balance consisting of two identical suspension systems were installed on each side of the tunnel to support vertically the two ends of a test section in the center of the tunnel's working section. The vertical and the torsional degrees-of-freedom were unrestrained. Each component of the force balance consisted of four vertical springs connected to flexure elements, which sensed the vertical motion. Strain gages mounted on the flexure elements were formed into Wheatstone bridges that measured the resistance due to change in strain. Downstream and upstream displacements were recorded separately by two channels. Each model was attached to the force balance at the two ends and end-plates were used to promote a two-dimensional flow around the model. A pitot tube was placed in the downstream region at a suitable distance from the centerline of a model to record the velocity fluctuations in the wake.

Section profiles of the six bluff sections considered in this study are shown in Fig. 1 and pertinent model properties are listed in Table 1. The twin-deck Fred Hartman bridge model was tested with 45.7 mm (representative of the gap/D ratio of the full-scale bridge)

and 30.5 mm gap. A single deck model of the bridge was also considered. The aeroelastic coefficients of the windward and the leeward sections of the Fred Hartman bridge were obtained under two different configurations. In one configuration, the leeward model was restrained while the response of the unrestrained windward model was recorded. In another configuration, the windward deck was restrained while the leeward deck was free to oscillate.

For each section model steady responses were obtained at a number of wind speeds before, inside, and beyond a lock-in region. Experiments were performed in both increasing and decreasing wind speeds. Two types of transient response behaviors were recorded – high initial displacement (higher than the steady response) to steady response and small initial displacement (smaller than the steady response) to steady response. In order to determine the Strouhal number of a model, the vertical and the rotational degrees of freedom of the model were restrained and velocity fluctuations in the wake were recorded using a pitot tube at several wind speeds. The natural frequency and the damping ratio of a model were obtained from its free-decay responses. More information on the experimental procedure and results can be found in Mashnad (2006).

Following important observations were made in the wind tunnel experiments:

- When a section model with two degrees-of-freedom experienced locked-in vibration, it was found to oscillate in a single mode only. When motion occurred in the vertical mode, the effect of the twisting motion on the aeroelastic forces acting on a section became negligibly small.

Table 1
Section model geometry and dynamic properties.

Properties	O	R	DIB	H	TB	FHB
Length (<i>L</i>), m	0.914	0.914	0.914	0.914	0.914	0.914
Depth (<i>D</i>), m	0.089	0.038	0.0476	0.0476	0.022	0.0287
Width (<i>B</i>), m	0.089	0.152	0.184	0.184	0.187	0.250
System mass (<i>M</i>), kg	2.4	2.3	1.9	1.9	2.7	1.9
Mass/unit length (<i>m</i>)	2.65	2.46	2.05	2.05	2.95	2.07
Width-to-Depth ratio (<i>B/D</i>)	–	4.00	3.87	3.87	8.5	8.7
Vertical frequency (<i>f_n</i>), Hz	3.33	3.45	3.72	3.7	3.15	5.0
Torsional frequency (<i>f_t</i>), Hz	–	6.32	5.50	5.5	6.19	6.26
Vertical damping ratio (<i>ζ_n</i>), %	0.19	0.26	0.24	0.24	0.24	0.17

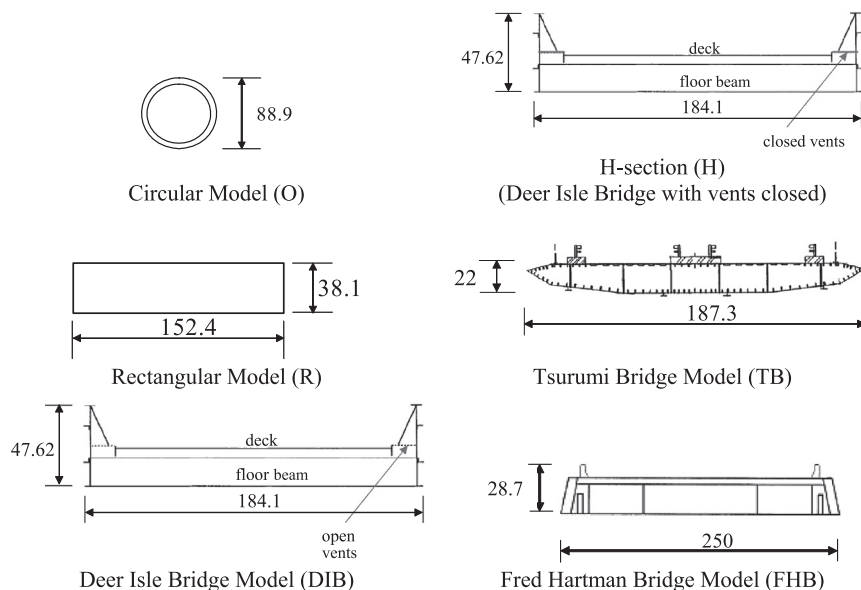


Fig. 1. Cross sections of six experimental models (all dimensions are in mm).

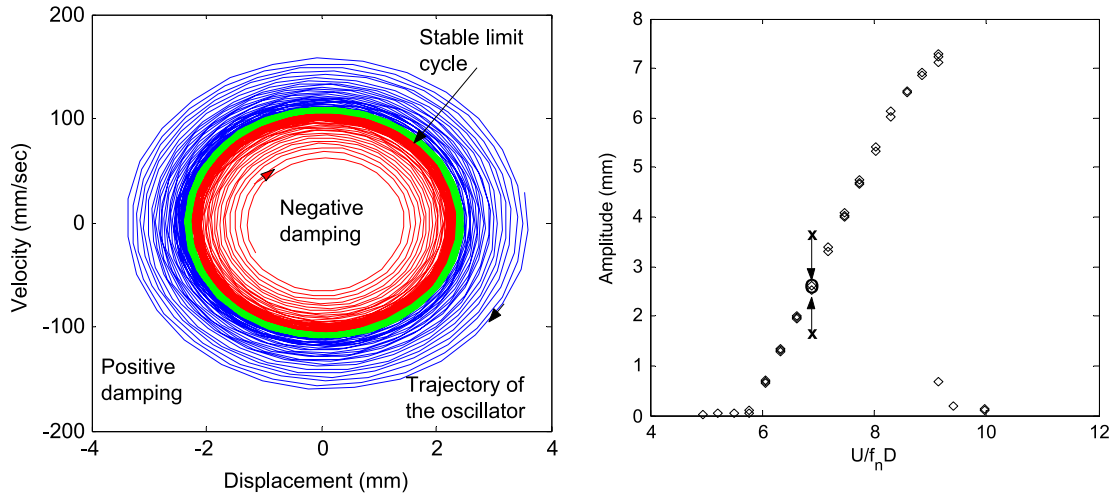


Fig. 2. Phase diagram and limit-cycle of an H section at a reduced wind speed of 6.9 (left) and corresponding initial and final positions of the oscillator on a response diagram (right).

- The decaying and the growing displacement responses of a model reached limit-cycle oscillations at lock-in wind speeds. As an example, the trajectories that reached a stable limit-cycle for the H section at a reduced wind speed of 6.9 are shown in a phase portrait and corresponding steady-state amplitudes versus reduced wind speeds are shown on a response diagram in Fig. 2. This type of behavior is observed in self-excited self-limiting nonlinear systems (Jordan and Smith, 1987). The location of a stable limit-cycle at a particular reduced wind speed depends on the mass and damping ratio of the oscillator in addition to the flow conditions.
- The total damping ratio of an oscillator in a flow during locked-in vibration demonstrated a nonlinear relationship with normalized amplitude. Instantaneous frequency analyses of the transient records by Wavelet Transform showed that the response frequency (response frequency of a section in lock-in regions) did not depend on the amplitudes of oscillation, but it depended weakly on reduced wind speeds. The natural frequency of an oscillator controls the frequency of oscillation of the fluid-structure system during locked-in vibration. The frequency of oscillation is approximately equal to the natural frequency of the oscillator.
- The vertical responses of all the sections have been combined in Fig. 3 for a relative comparison. The circular section exhibited one lock-in region, while the rectangular, H-shape, and Deer Isle bridge sections exhibited two lock-in regions. The streamlined Tsurumi bridge exhibited one lock-in region. The Fred Hartman bridge section experienced three lock-in regions for single deck and for the configuration where the windward deck was free to oscillate and the leeward deck was fixed (Fig. 3). In another configuration, where the windward deck was fixed and the leeward deck was free, the leeward deck exhibited only one lock-in instability at wind speeds that corresponded to the second lock-in instability for the former two configurations. It should be noted that these configurations are idealized versions of the actual structural system of the prototype bridge. The towers of the prototype bridge are integrally joined with the two decks at the deck level and, as a result, the movement of one deck affects the response of the other.

3. Proposed mathematical model

Locked-in response of a full-scale bridge is analyzed in two steps. Firstly, lock-in wind speeds are estimated from those identified in a wind tunnel experiment on a spring-mounted section

model of a bridge. Secondly, the response amplitudes at each lock-in region of the prototype bridge are estimated using an amplitude-dependent aeroelastic equation of motion.

3.1. Identification of lock-in regions

The wind speeds for onset and end of locked-in vibration of a full-scale bridge can be obtained from wind tunnel tests using Eq. (1) based on the assumptions that response in each mode is separable and that the Strouhal numbers for full-scale structure and test model are identical.

$$U = U_r f_n D \quad (1)$$

where, U_r is the reduced wind speed of locked-in vibration determined in wind tunnel experiments, f_n is the n -th modal frequency of full-scale bridge, and D is a characteristic dimension of full-scale deck. This equation cannot be used for lock-in wind speeds as the Strouhal relationship does not hold true during locked-in vibration.

3.2. Equation of motion

The general form of an equation of motion that describes the across-flow response of a single-degree-of-freedom spring-mounted rigid cylinder due to self-excited or motion-induced fluid forces can be expressed as

$$m(\ddot{h} + 2\zeta_n \omega_n \dot{h} + \omega_n^2 h) = F(h, \dot{h}, \ddot{h}, U, t) \quad (2)$$

where, m is the mass/length of cylinder, h is the structural response and over dot represents derivatives of h with respect to time, ω_n is the n -th natural frequency, ζ_n is the mechanical damping ratio, U is the mean free-stream velocity, and F is the combined general and motion-induced fluid force per unit span. In the mathematical models for flutter, an instability also caused by motion-induced fluid forces, the aeroelastic forces are expressed in terms of the motion of a structure. The flutter model for vertical motion of a spring-mounted rigid section with vertical and rotational degrees-of-freedom has the well-known form (Scanlan and Tomko, 1971; Scanlan and Jones, 1990)

$$m(\ddot{h} + 2\zeta_n \omega_n \dot{h} + \omega_n^2 h) = \frac{1}{2} \rho U^2 B \left[KH_1^*(K) \frac{\dot{h}}{U} + KH_2^*(K) \frac{B \dot{\alpha}}{U} + K^2 H_3^*(K) \alpha + K^2 H_4^*(K) \frac{h}{B} \right] \quad (3)$$

where, h is the vertical displacement, α is the angle of twist in radians, ρ is the unit mass of air, K is the $B\omega/U$, ω is the circular response

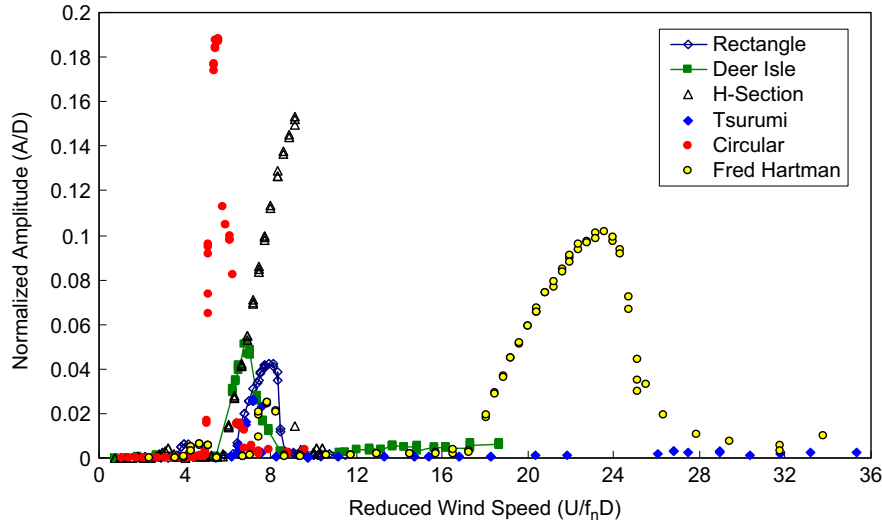


Fig. 3. Normalized amplitude of vertical oscillations of six models. For Fred Hartman bridge model two decks placed side-by-side with 46.7 mm gap ($\text{gap}/D=1.63$) and leeward deck restrained.

frequency of the model, B is the model width, and H_1^* to H_4^* are the empirical aeroelastic coefficients also called flutter derivatives.

As this model is widely used for predicting flutter instability of long-span bridges, the analysis method and associated experimental procedure are quite familiar to many in the bridge engineering community. Therefore, a similar format for modeling locked-in vibration was adopted herein incorporating the main features observed in wind tunnel experiments. In this study, only vertical motion is considered based on the experimental observation that a spring-mounted model oscillates in single mode (either vertical or torsional mode) at locked-in vibration, the following revised form of the original flutter equation is proposed to model the locked-in vibration of an oscillator:

$$m(\ddot{h} + 2\zeta_n \omega_n \dot{h} + \omega_n^2 h) = \frac{1}{2} \rho U^2 B \left[K H_1^*(A/D, K) \frac{\dot{h}}{U} + K^2 H_4^*(K) \frac{h}{D} + C_L(K) \sin(\omega_s t + \phi) \right] \quad (4)$$

where, C_L is the aerodynamic lift coefficient, ω_s is the shedding frequency, and ϕ is the phase angle. Experimental results demonstrated that H_1^* is a function of both normalized amplitudes (A/D) and reduced wind speeds ($U/f_n D$), and H_4^* is of reduced wind speeds ($U/f_n D$). The third forcing term in Eq. (4) represents direct aerodynamic force, which is negligibly small compared with the aeroelastic forces at lock-in and therefore can be ignored (Scanlan, 1998; Simiu and Scanlan, 1986). It should be pointed out that H_1^* in Eq. (4) represents a motion-induced nonlinear aeroelastic parameter whereas the H_1^* in Scanlan's flutter equation (Eq. (3)) does not depend on motion. Depending on the initial condition of an oscillator, the aeroelastic damping component of Eq. (4) either adds or removes energy from the dynamic system and once the net damping in the system equals zero, the oscillator reaches a steady-state response. The amplitude dependent H_1^* serves similar to the nonlinear damping component of the Van der Pol equation. Eq. (4) cannot be used to model locked-in vibration if H_1^* does not vary with structural motion.

3.2.1. Extraction of aeroelastic coefficients

To identify the aeroelastic coefficients from experimental time-histories, Eq. (4) needs to be restructured as follows:

$$\ddot{h} + \left(2\zeta_n \omega_n - \frac{1}{2} \frac{\rho B D}{m} \omega H_1^* \right) \dot{h} + \left(\omega_n^2 - \frac{1}{2} \frac{\rho B D}{m} \omega^2 H_4^* \right) h = 0 \quad (5)$$

Eq. (5) can be recognized as the equation of motion of an equivalent free-vibration system in a flow. This formula can be expressed in terms of total damping, γ , and response frequency, ω , as follows:

$$\ddot{h} + 2\gamma \omega \dot{h} + \omega^2 h = 0; \text{ where } 2\omega\gamma = 2\zeta_n \omega_n - \frac{1}{2} \frac{\rho B D \omega}{m} H_1^*; \quad \omega^2 = \omega_n^2 - \frac{1}{2} \frac{\rho B D}{m} \omega^2 H_4^* \quad (6a, b, c)$$

After rearranging the terms of Eqs. (6b) and (6c), H_1^* and H_4^* can be expressed as follows:

$$H_1^*(A/D, K) = \frac{4\zeta_n m}{\rho B D} \left[\frac{\omega_n}{\omega} - \frac{\gamma}{\zeta_n} \right]; \quad H_4^*(K) = \frac{2m}{\rho B D} \left[\left(\frac{\omega_n}{\omega} \right)^2 - 1 \right] \quad (7a, b)$$

H_1^* is a function of the total damping ratio, γ , which depends on amplitude of oscillation. The term $\zeta_n m / \rho B D$ in Eq. (7a) is a mass-damping parameter for a section, also known as Scruton number. To identify the aeroelastic coefficients for a section model, it is necessary to determine the total damping ratio, γ , and the circular response frequency, ω , from wind tunnel experiments at several wind speeds. Using the Hilbert Transform method, the envelope of a decaying or a growing displacement signal at a particular wind speed is obtained. After converting the envelope to natural logarithmic form, the envelope is divided into small segments - usually one to two response periods ($1/f_n$). The total damping ratio, γ , is estimated by fitting a least-squares line through the data points in each segment. The slope of the fitted line is $-2\pi f_n \gamma$ at a normalized amplitude, A/D , at the center of the segment. The system properties ζ_n and ω_n of the same section can be obtained from free-decay tests in still air. Finally, H_1^* and H_4^* can be evaluated using Eqs. (7a) and (7b), respectively.

3.2.2. Aeroelastic coefficients H_1^* and H_4^*

Fig. 4 presents H_1^* as a function of normalized amplitudes for the H section at two reduced velocities within the lock-in region that ranged from $U/f_n D = 6.5$ to 8.7. The point of intersection of the horizontal and the vertical lines on a curve indicates the equilibrium state where energy imparted to the system is completely balanced by the inherent damping in the system. The value of A/D at this intersection corresponds to the steady-state response of the oscillator. The segment of the H_1^* versus A/D curve above the horizontal line corresponds to a growing response that was obtained by applying a small initial displacement to the section

model. The lower segment, below the horizontal line, of the curve corresponds to a decaying response. The combined curve fully describes the growing and decaying trajectories of the oscillator to

a limit-cycle response during locked-in vibration at a particular wind speed. The two curves in Fig. 4 obtained at two wind speeds indicate that the relationship between A/D and H_1^* is nonlinear and

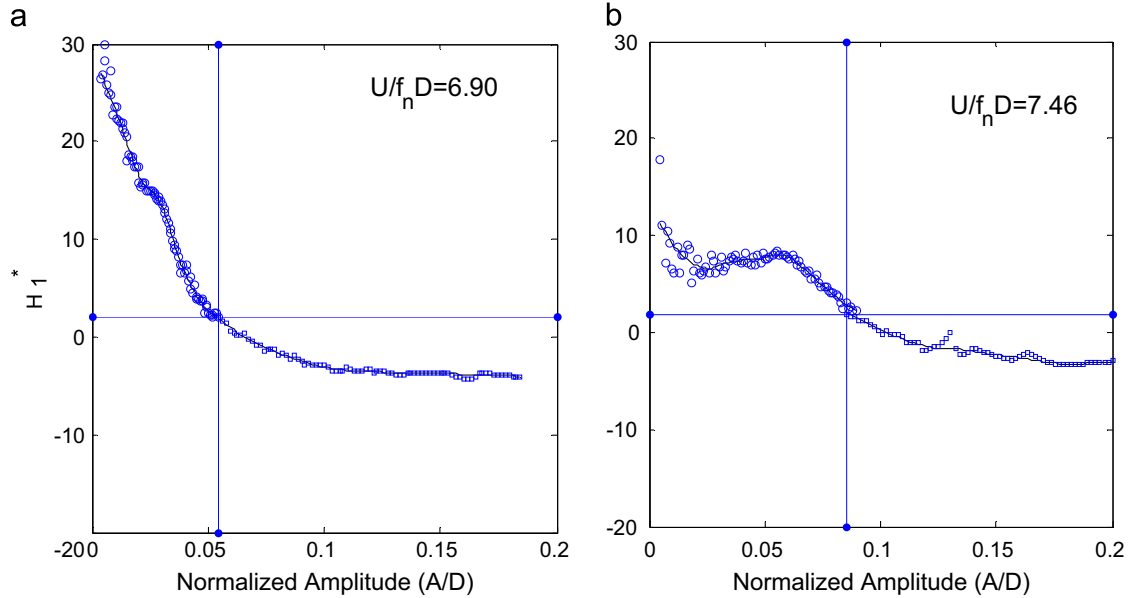


Fig. 4. H_1^* as function of normalized amplitude for H section at several reduced wind speeds.

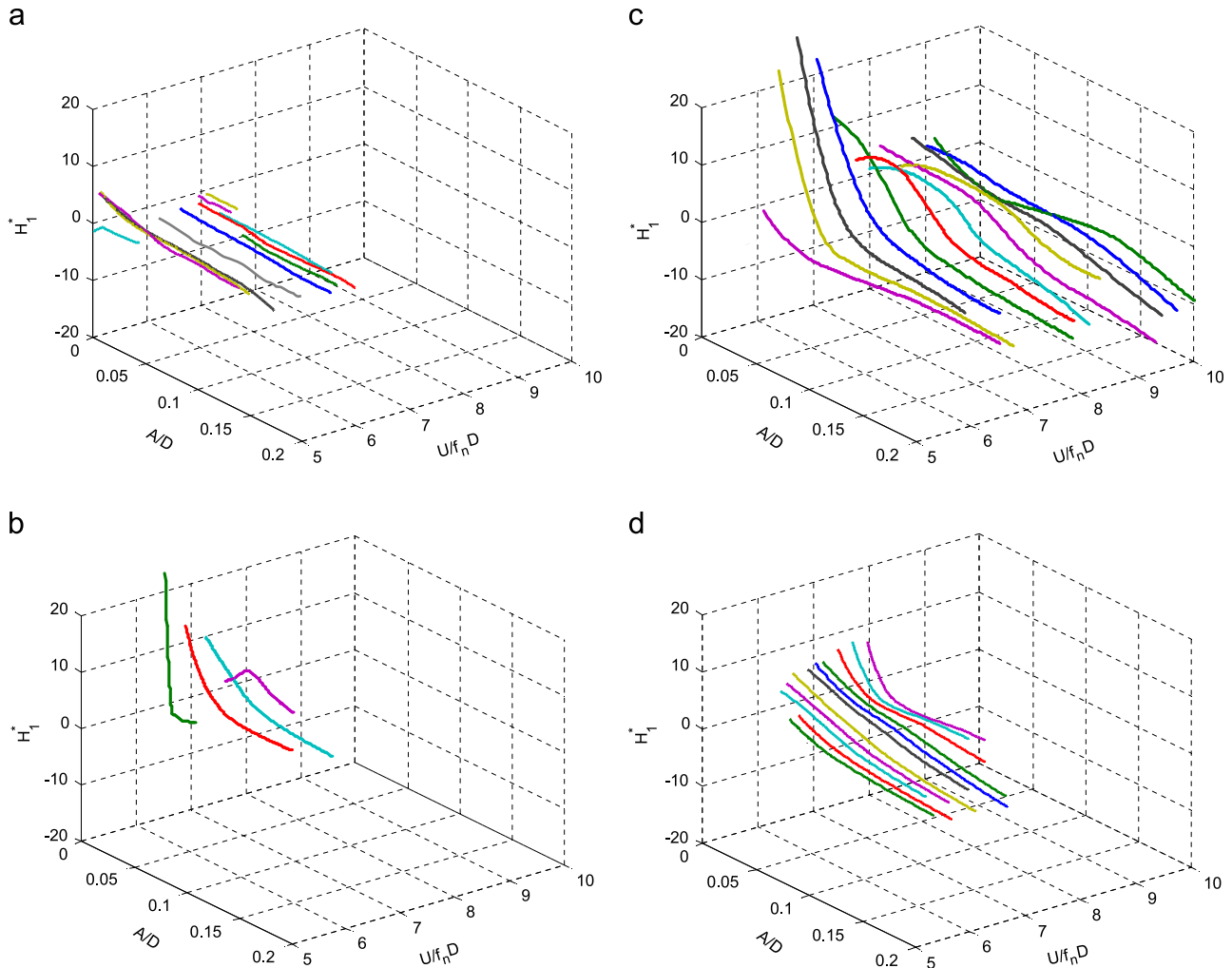


Fig. 5. H_1^* at lock-in regions for (a) circular section at $U/f_n D=5-6.4$ (b) Tsurumi Bridge section at $U/f_n D=6.1-8.3$ (c) H-section at $U/f_n D=5.8-9.4$, and (d) Deer Isle bridge section at $U/f_n D=5.8-8.5$.

amplitude dependent. The shapes of the two curves are different, which indicates that the detailed characteristics of the fluid-structure interaction are different at different reduced wind speeds. The scatter in H_1^* coefficients at low A/D in Fig. 4 is caused by the undulating response envelope at the initial part of the time history due to the presence of the Strouhal frequency before a complete lock-in state was established. The characteristics of the H_1^* versus A/D curves also vary for different section profiles. Fig. 5 presents the H_1^* coefficients as a function of reduced wind speed and normalized amplitude for four sections inside the lock-in regions. Fig. 6 presents H_1^* for the Fred Hartman bridge model for single deck and twin deck configurations.

The H_4^* coefficients for the rectangular and the H sections are grouped in Fig. 7. It was observed that the circular and the rectangular sections have small values of H_4^* at the beginning of a lock-in region followed by larger values at the end of the lock-in region. On the other hand, the H and the Deer Isle bridge sections demonstrated a different trend where H_4^* continued to decrease from the beginning to the end of the lock-in region. In most cases, the response frequency tends to increase slightly at higher reduced velocities, which resulted in a decreasing trend in H_4^* . Negative values of H_4^* indicate a stiffening effect whereas positive values indicate a softening effect. It should be emphasized that if the frequency resolution of a response spectrum is not sufficient, such small differences may not be captured. Small differences in frequency

can result in large differences in H_4^* as the coefficient is obtained by the multiplication of several large parameters. The H_4^* coefficients for the Fred Hartman bridge models tested at several configurations are shown in Fig. 8. The general forms of the curves are found similar. The aeroelastic coefficients are positive up to a reduced wind speed of about 23 beyond which the coefficients become negative.

4. Full-scale structural response analysis

The locked-in response of a full-scale structure can be determined using the non-dimensional aeroelastic coefficients H_1^* and H_4^* presented in the previous section in a numerical procedure consistent with Scanlan and Jones (1990). As the formula of Eq. (4) was developed for a spring-mounted rigid model oscillating in a nominally two-dimensional flow, the formula needs to be modified to accommodate the characteristics of a long-span, flexible structure. A full-scale structure has an infinite number of modes of vibration and response amplitudes vary along the span. As the amplitudes of vibration vary, the amplitude dependent aeroelastic forces on the structure will also vary along the span. The aeroelastic forces may also not be perfectly correlated along the span due to loss of spanwise correlation.

The analysis of a long-span bridge considers standard vibration modal analysis, where the structural response is represented by h

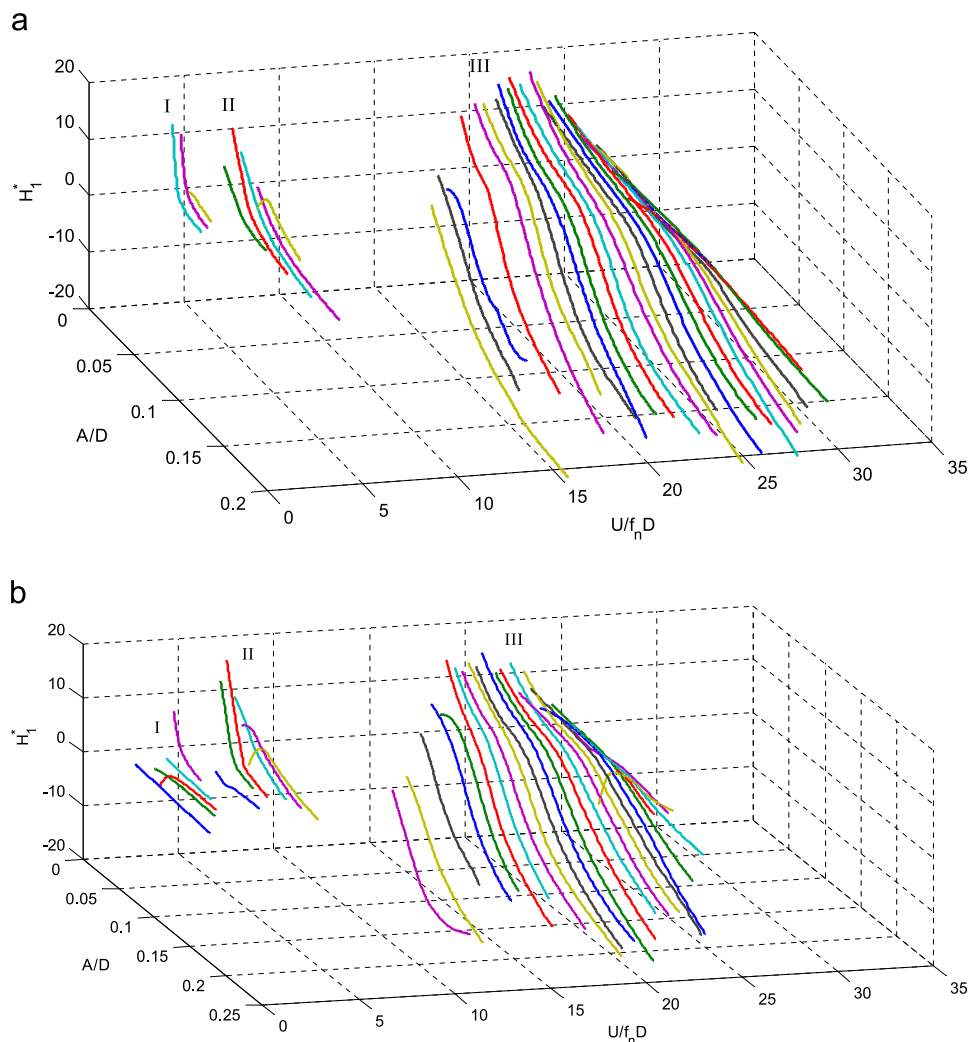


Fig. 6. H_1^* of Fred Hartman bridge for (a) single deck, and (b) windward deck free and leeward deck restrained, gap/ $D=1.64$ (represents full-scale gap). I: first lock-in, II: second lock-in, and III: third lock-in regions.

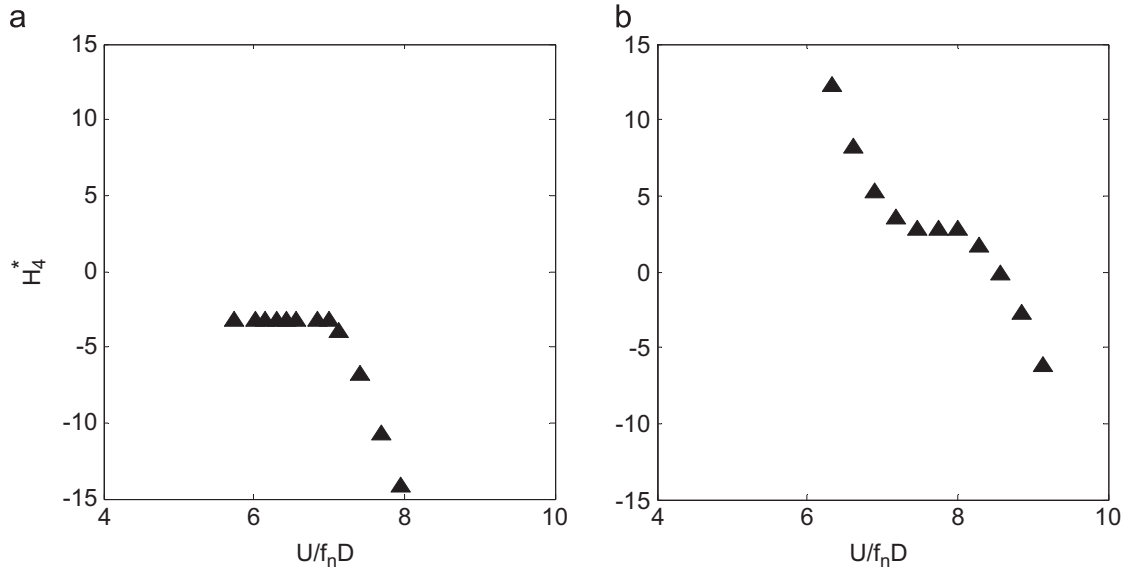


Fig. 7. H_4^* for (a) rectangular section and (b) H section. Coefficients correspond to second lock-in region.

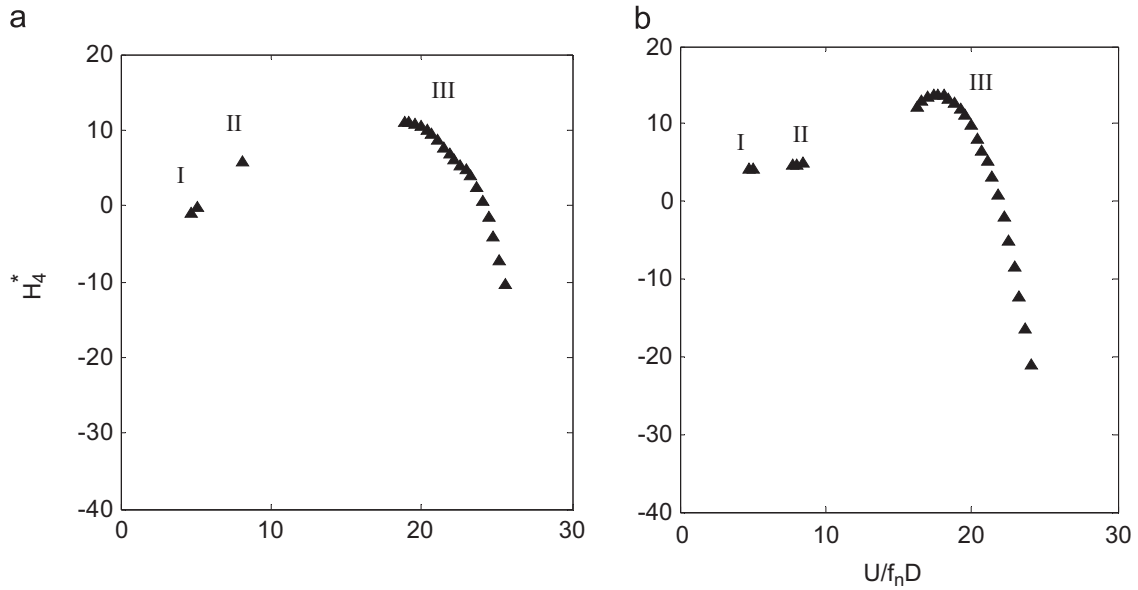


Fig. 8. H_4^* for Fred Hartman bridge model. (a) Single deck, (b) double deck configuration, upstream deck free, gap/ $D=1.64$. I: first lock-in, II: second lock-in, and III: third lock-in regions.

(x, t) as follows (Scanlan and Jones, 1990):

$$h(x, t) = \sum_n \phi_n(x) B \eta_n(t) \quad (8)$$

where, x is the coordinate along bridge span; $\phi_n(x)$ is the dimensionless n -th vertical mode shapes; $\eta_n(t)$ is the generalized coordinate of the mode; B is the width of deck. If the natural frequency of the n -th mode is ω_n and damping ratio ζ_n , the motion $\eta_n(t)$ can be expressed as

$$M_n (\ddot{\eta}_n + 2\zeta_n \omega_n \dot{\eta}_n + \omega_n^2 \eta_n) = Q_n(t);$$

$$\text{where } M_n = \int_{\text{Structure}} \psi_n^2(x, y, z) dm(x, y, z) \quad (9a, b)$$

where ψ_n is the n -th mode shape of a bridge, dm is the unit mass, and $Q_n(t)$ is the generalized force due to vortex shedding in this study, and M_n is the generalized inertia of the n -th mode.

The lift force on a bridge deck consists of aeroelastic and buffeting force components. During locked-in vibration, the buffeting force is assumed to become negligibly small compared to aeroelastic force (Ehsan, 1989). Additionally, full-scale measurements and wind tunnel experiments strongly suggest that a structure oscillates in only one mode at a time when locked-in vibration occurs. Therefore, a single-mode dynamic analysis is generally considered (Ehsan, 1989; Macdonald et al., 2002; Irwin, 1998). Based on these observations, the formula for generalized vortex-induced force can be expressed as

$$\ddot{\eta}_n + 2\zeta_n \omega_n \dot{\eta}_n + \omega_n^2 \eta_n = \frac{1}{2} \frac{\rho U^2 B}{M_n} \left[\int_0^L K H_1^* B^2 \phi_n^2 dx \frac{\dot{\eta}_n}{U} + \int_0^L K^2 H_4^* B^2 \phi_n^2 dx \frac{\eta_n}{D} \right] \quad (10)$$

where, L is the bridge length. For a twin deck bridge, Eq. (10) can be modified to account for the different aeroelastic coefficients of

windward and leeward decks as follows:

$$\ddot{\eta}_n + 2\zeta_n \omega_n \dot{\eta}_n + \omega_n^2 \eta_n = \frac{1}{2} \frac{\rho U^2 B}{M_n} \left[\left(\int_0^L K H_{1(wind)}^* B^2 \phi_n^2 dx + \int_0^L K H_{1(lee)}^* B^2 \phi_n^2 dx \right) \frac{\dot{\eta}_n}{U} + \left(\int_0^L K^2 H_{4(wind)}^* B^2 \phi_n^2 dx + \int_0^L K^2 H_{4(lee)}^* B^2 \phi_n^2 dx \right) \frac{\eta_n}{D} \right] \quad (11)$$

4.1. Full-scale structural response considering loss of spanwise correlation

Flows encountered on prototype structures have three-dimensional characteristics (Davenport, 1962; Simiu and Scanlan, 1986). Turbulence in flow and the variation in the geometries of a structure also necessitate the consideration of three-dimensional properties of flow. The correlation of the aeroelastic forces will decrease when the separation distance between two points on a bridge increases. Correlation will also decrease due to turbulence in on-coming wind and spatial variation of mean wind velocity along the span. On the other hand, correlation tends to increase in locked-in vibration with increased amplitude of vibration. The randomness of the spanwise force components can be represented by a narrow-band stationary random process. The integral of the cross-spectral density function, $S_{FF}(x_A, x_B, K)$, of vortex-induced lift force, $F(x, t)$, is related to the average of the square of the force per unit length as (Blevins, 1990)

$$\int_0^\infty S_{FF}(x_A, x_B, K) dK = \overline{[F(x)]^2}; \quad \text{where } \overline{[F(x)]^2} = g(x_A)g(x_B)f(c, |x_A - x_B|/L) \quad (12a, b)$$

where, x is the $|x_A - x_B|$; $g(x_A)$ is the square root of the mean square lift forces at spanwise location x_A ; $f_c(c, |x_A - x_B|/L)$ is the spanwise correlation function whose form is yet to be determined. The over-bar denotes the time average of the force. To account for the loss of correlation, the force component associated with H_1^* of Eq. (10) can be modified in line with Eq. (12) as follows:

$$\bar{I}_{1n} = \chi \left[\int_0^L \int_0^L |H_1^*(x_A)H_1^*(x_B)| B^2 \phi_n^2(x_A) B^2 \phi_n^2(x_B) f(c, |x_A - x_B|/L) dx_A dx_B \right]^{\frac{1}{2}} \quad (13)$$

Similarly the force component associated with H_4^* in Eq. (10) can be expressed as

$$\bar{I}_{4n} = \chi \left[\int_0^L \int_0^L |H_4^*(x_A)H_4^*(x_B)| B^2 \phi_n^2(x_A) B^2 \phi_n^2(x_B) f(c, |x_A - x_B|/L) dx_A dx_B \right]^{\frac{1}{2}} \quad (14)$$

where χ is the sign of the integral inside the square root. $\bar{I}_{1n}$ and $\bar{I}_{4n}$ are the modified aeroelastic damping and aeroelastic stiffness parameters, respectively, that consider loss of spanwise correlation. Using these terms, the equation of motion of a bridge can be expressed as

$$\ddot{\eta}_n + 2\zeta_n \omega_n \dot{\eta}_n + \omega_n^2 \eta_n = \frac{1}{2} \frac{\rho U^2 B}{M_n} \left[K \bar{I}_{1n} \frac{\dot{\eta}_n}{U} + K^2 \bar{I}_{4n} \frac{\eta_n}{D} \right] \quad (15)$$

For a twin deck bridge, the equation of motion can be expressed as

$$\ddot{\eta}_n + 2\zeta_n \omega_n \dot{\eta}_n + \omega_n^2 \eta_n = \frac{1}{2} \frac{\rho U^2 B}{M_n} \left[K \bar{I}_{1n(wind)} \frac{\dot{\eta}_n}{U} + K \bar{I}_{1n(lee)} \frac{\dot{\eta}_n}{U} + K^2 \bar{I}_{4n(wind)} \frac{\eta_n}{D} + K^2 \bar{I}_{4n(lee)} \frac{\eta_n}{D} \right] \quad (16)$$

Following change of variables is made to simplify numerical computation of Eqs. (13) and (14):

$$\frac{x_A}{L} = \mu; \quad dx_A = L d\mu; \quad 0 \leq \mu \leq 1 \quad (17)$$

$$\frac{x_B}{L} = \lambda; \quad dx_B = L d\lambda; \quad 0 \leq \lambda \leq 1 \quad (18)$$

Eqs. (17) and (18) are substituted in Eqs. (13) and (14) to obtain

$$\bar{I}_{1n} = \chi L B^2 \left[\int_0^1 \int_0^1 |H_1^*(K, A/D(\mu))H_1^*(K, A/D(\lambda))| \phi_n^2(\mu) \phi_n^2(\lambda) f_c(c, |\mu - \lambda|) d\mu d\lambda \right]^{\frac{1}{2}} \quad (19)$$

$$\bar{I}_{4n} = \chi L B^2 \left[\int_0^1 \int_0^1 |H_4^*(K)|^2 \phi_n^2(\mu) \phi_n^2(\lambda) f_c(c, |\mu - \lambda|) d\mu d\lambda \right]^{\frac{1}{2}} \quad (20)$$

4.2. Models for spanwise loss of correlation

The three-dimensional effects of wind on a structure can be modeled by the loss of correlation of pressure between two spanwise points based on wind tunnel experiments or full-scale studies (Vickery, 1966; Ehsan, 1989; Matsumoto et al., 2003). As the variation of the aeroelastic forces along the span of a full-scale bridge is not fully known due to lack of study of this topic, generally the correlation patterns of lateral wind are considered as a basis. In buffeting analysis of long-span bridges, to describe the loss of correlation between wind speeds at spanwise points x_A and x_B , the following function is often considered (Simiu and Scanlan, 1986):

$$f_c = e^{-c|x_A - x_B|/L} \quad \text{where} \quad \frac{8nL}{U} \leq c \leq \frac{16nL}{U} \quad (21)$$

where n is the turbulence frequency in Hz, L is the length of structure, and U is the mean wind speed.

Wind tunnel experiments demonstrate that correlation of the aeroelastic forces depends on amplitude of oscillation as well as spanwise separation during locked-in vibration of spring-mounted rigid section models. Ehsan (1989) proposed a general normalized amplitude-dependent correlation function based on pressure fluctuations in the wake of a freely oscillating H-shape section in a wind tunnel of the form

$$f_c = \exp\left(\frac{-a}{b + A/D} \sqrt{|x_B - x_A|/D}\right) \quad (22)$$

The experimental values of a and b were 0.0079 and 0.0135, respectively (Ehsan, 1989). These two correlation functions were incorporated in Eqs. (19) and (20) for modeling loss of correlation of the aeroelastic forces.

5. Full-scale locked-in response of Fred Hartman bridge

5.1. Measured full-scale bridge response

The twin-deck Fred Hartman bridge has a main span of 381 m and two side-spans of 147 m. Each deck is composed of precast concrete slabs on steel girders and is supported by 96 cables attached to a diamond-shaped tower located at each end of the main span. Each deck is 25 m wide and 2.87 m deep (bottom of bottom flange of main girder to top of concrete barrier). A monitoring system installed on the bridge in 1997 recorded cable and deck accelerations, cable displacements, wind speeds/directions and other relevant meteorological information for over five years (Main and Jones, 2001; Ozkan, 2003).

Fig. 9 presents representative root mean square (RMS) amplitudes of vertical response of the bridge measured at midspan and at midpoint between the midspan and the south tower. Vortex-induced vibration was observed in four regions which are labeled as Region-I, -II, -III and -IV. The first three modes of vibration and damping ratio obtained by a system-identification technique from the full-scale data are provided in Table 2. Due to twin deck configuration, two frequencies exist for a given mode shape. The

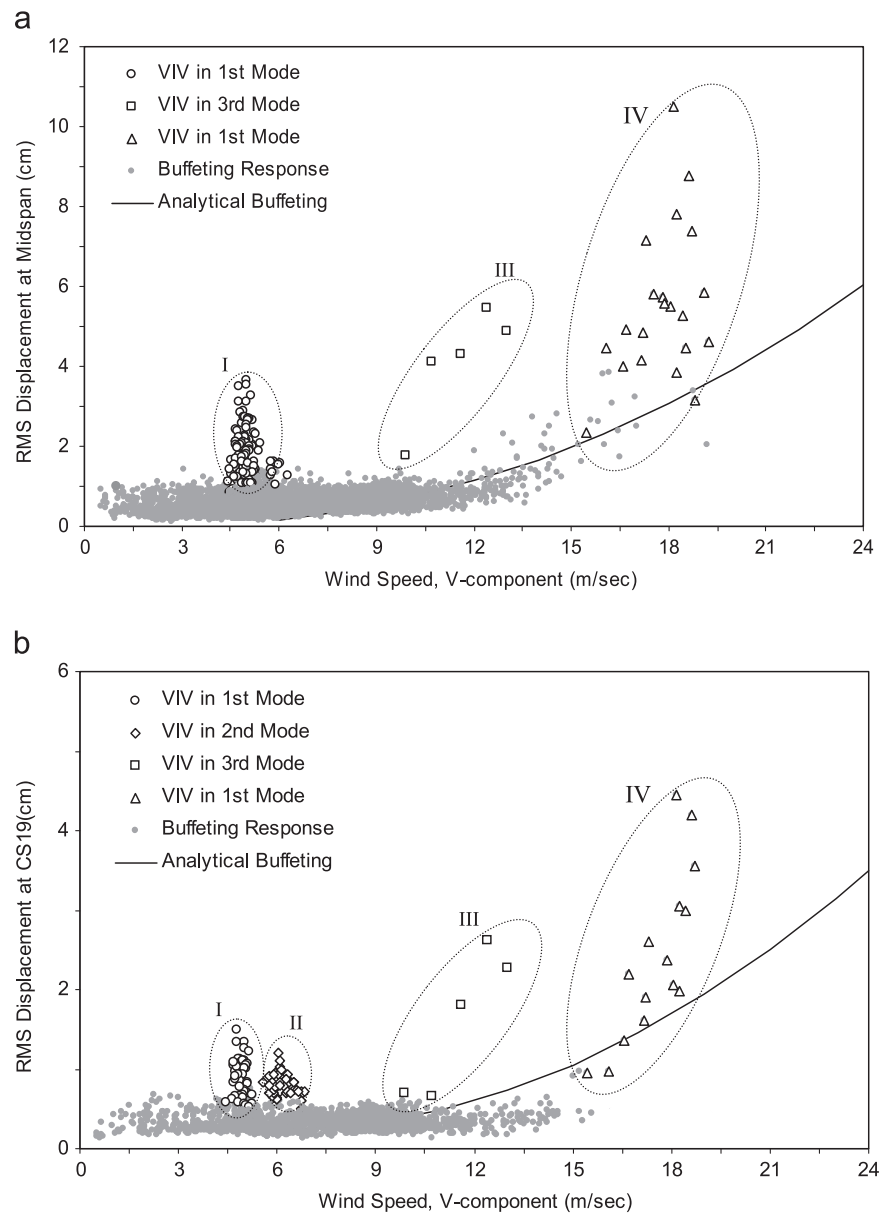


Fig. 9. Full-scale vortex-induced and buffeting response at (a) midspan, and (b) midpoint between the midspan and the south tower. I: first lock-in, II: second lock-in, and III: third lock-in regions.

Table 2
Modal frequencies and damping ratios of the Fred Hartman Bridge obtained from full-scale measurements.

Bridge mode	Phase	Median frequency (Hz)	Semi-interquartile frequency (Hz)	Median damping (%)	Semi-interquartile damping ratio (%)
1st Vertical	In	0.287	0.285–0.288	0.70	0.61–0.77
1st Vertical	Out	0.301	0.298–0.302	0.70	0.61–0.77
2nd Vertical	In	0.371	0.370–0.372	0.54	0.46–0.59
2nd Vertical	Out	0.382	0.381–0.383	0.54	0.46–0.59
3rd Vertical	In	0.567	0.566–0.567	0.41	0.38–0.45
3rd Vertical	Out	0.584	0.583–0.585	0.41	0.38–0.45

first frequency corresponds to in-phase motion where both decks deform in the same direction and the second frequency represents out-of-phase motion where one deck deforms in some direction while the second deck deforms in the opposite direction. Detailed information on system identification and full-scale response study can be found in (Mashnad and Jones, 2005; Mashnad, 2006).

5.2. Estimated lock-in regions for full-scale bridge

Three lock-in regions were observed in wind tunnel tests of the Fred Hartman bridge section model. Using Eq. (1), three lock-in regions associated with each mode of the full-scale bridge were determined from the experimental results. The predicted lock-in regions for the first three vertical modes, each with in-phase and

out-of-phase modes, of the bridge decks are listed in Table 3. For example, the table shows that bridge deck will potentially experience locked-in oscillations in the first in-phase vertical mode (0.287 Hz) in three bands of wind speeds: 3.2–4.5 m/sec, 5.4–7.1 m/sec, and 13.7–22.2 m/sec. For the first out-of-phase vertical mode (0.301 Hz), these bands are 3.3–4.8 m/sec, 5.7–7.4 m/sec, and 14.4–23.3 m/sec. It is observed that the regions where large-amplitude oscillations occurred on the bridge (Fig. 9) closely matched the predicted lock-in regions.

5.3. Numerical response of full-scale bridge

Response history of the Fred Hartman bridge deck was calculated by the fourth-order Runge–Kutta method using a state-space representation. This numerical algorithm is considered more accurate and stable (Meirovitch, 1986). The formula developed in this study was solved by an iterative scheme because the aeroelastic forces change at every time step due to amplitude dependent property of H_1^* . For each simulation, the initial displacement was chosen such that it fell within the range of A/D obtained from wind tunnel tests. The initial structural velocity was set to zero. Simulation continued until the aeroelastic damping ratio equaled the structural damping ratio and structural motion reached steady-state amplitudes. To account for spanwise loss of correlation of the aeroelastic forces, locked-in response of the bridge was simulated using correlation Eq. (21) for several values of c and several damping ratios for each c . On the other hand, the parameters a and b of the amplitude-dependent correlation function (Eq. (22)) were estimated by fitting simulated response amplitude to measured full-scale response amplitudes using a regression analysis. The values of a and b estimated by Ehsan (1989) were not used in the analysis because those were obtained for an H-shaped section in a wind tunnel and, therefore, are not representative of

actual field condition. A different set of a and b was calculated for each lock-in regions observed in Fig. 9. The aeroelastic coefficients extracted for the windward section were used for both bridge decks in one set of analyses. This set of the aeroelastic coefficients will be called Set-I. In the second set of numerical simulations, the aeroelastic coefficients obtained from the windward section were applied to the windward bridge deck and those obtained from the leeward section were applied on the leeward bridge deck. This combination of the aeroelastic coefficients will be called Set-II.

Analysis was performed using MATLAB 6.5. A three-dimensional finite element (FE) model of the bridge was used to determine structural frequencies and mode shapes. Calculated frequencies matched well the measured frequencies of the bridge. Damping ratios obtained from full-scale data were used in numerical simulation.

5.3.1. Numerical response obtained from set-I aeroelastic coefficients

Primary locked-in response was observed on the bridge at wind speeds of 15–20 m/s in the first in-phase vertical mode (Region-IV in Fig. 9a). The simulated RMS displacement amplitudes at midspan in the first vertical mode are compared with the measured full-scale responses in Fig. 10a. Responses were calculated for several damping ratio. The results indicate that 0.70% damping, which is the measured median damping ratio in the first vertical mode, with $c=3$ resulted in amplitudes that closely matched the maximum measured response amplitude of 10.5 cm. The response amplitudes obtained from the amplitude-dependent correlation function demonstrated good agreement with measured response for $a=0.004$ and $b=0.027$ (Fig. 10b).

The simulated and the measured responses in the first vertical mode of the bridge at wind speeds of 4.5–6.5 m/sec (Region-I in Fig. 9a) is shown in Fig. 11(a). The maximum RMS displacement in Region-I was measured 3.6 cm at a damping ratio of 0.70% which agreed well with the simulated response with 0.70% damping ratio and $c=3$. The response obtained from the amplitude-dependent correlation function demonstrated good agreement with measured response for a damping ratio of 0.70% with $a=0.00316$ and $b=0.027$ (Fig. 11b).

The full-scale response of Region-II (Fig. 9b) is associated with the locked-in response of second vertical mode of the bridge. It is observed in Fig. 12 that the simulated responses at 0.54% damping, which is the measured damping on the full-scale structure, were much higher than the full-scale measured response even for $c=5$ (a large value in consideration of locked-in response). Given the fact that the full-scale responses in the other three regions (Region-I, -III and -IV) were well predicted by this model, it can be suggested that the response in Region-II did not attain its

Table 3
Estimated lock-in regions for full-scale Fred Hartman bridge. Two decks placed side-by-side in a wind tunnel with gap/ $D=1.64$ and downstream deck fixed.

Lock-in regions	In-phase modes						Out-of-phase modes					
	$f_1=0.287$		$f_2=0.371$		$f_3=0.567$		$f_1=0.301$		$f_2=0.382$		$f_3=0.584$	
	U_o	U_e	U_o	U_e	U_o	U_e	U_o	U_e	U_o	U_e	U_o	U_e
1st	3.2	4.5	4.1	5.9	6.3	9.0	3.3	4.8	4.2	6.1	6.5	9.3
2nd	5.4	7.1	7.0	9.2	10.8	14.1	5.7	7.4	7.3	9.5	11.1	14.5
3rd	13.7	22.2	17.8	28.8	27.3	44.2	14.4	23.3	18.4	29.8	28.2	45.6

Subscripts 'o' stands for onset and 'e' for ending wind speed.

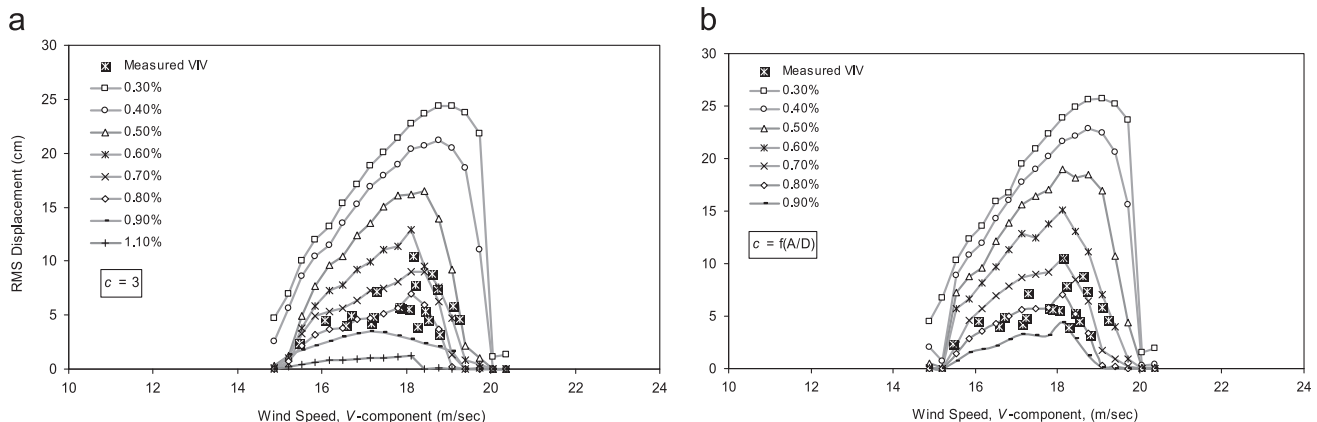


Fig. 10. Comparison between measured and simulated responses at midspan in first vertical mode of region-IV at several damping ratio. (a) $c=3$ (b) c function of A/D .

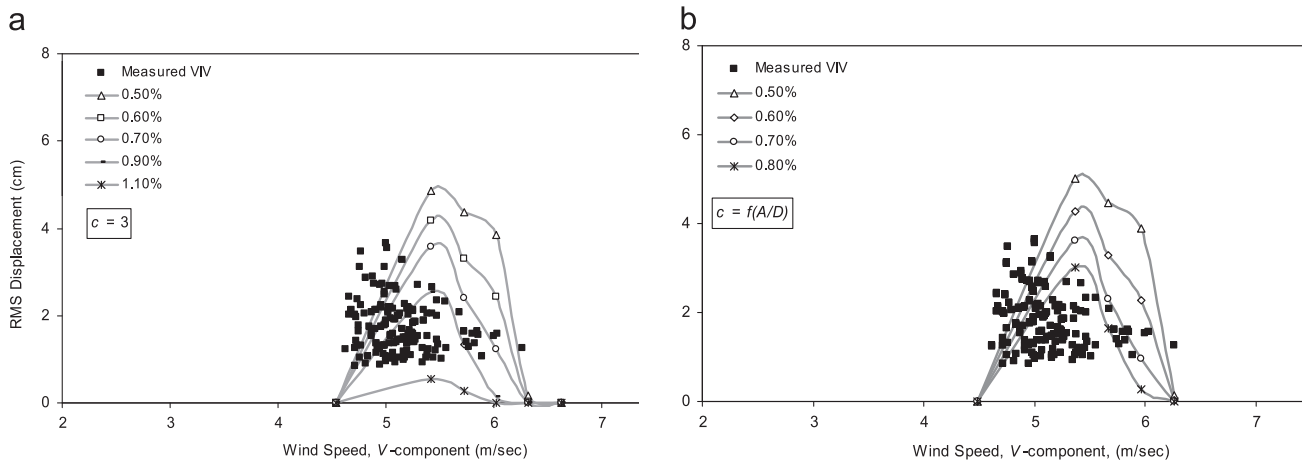


Fig. 11. Comparison between measured and simulated response at midspan in first vertical mode corresponding to region-I at several damping ratio. (a) $c=3$ and (b) c function of A/D .

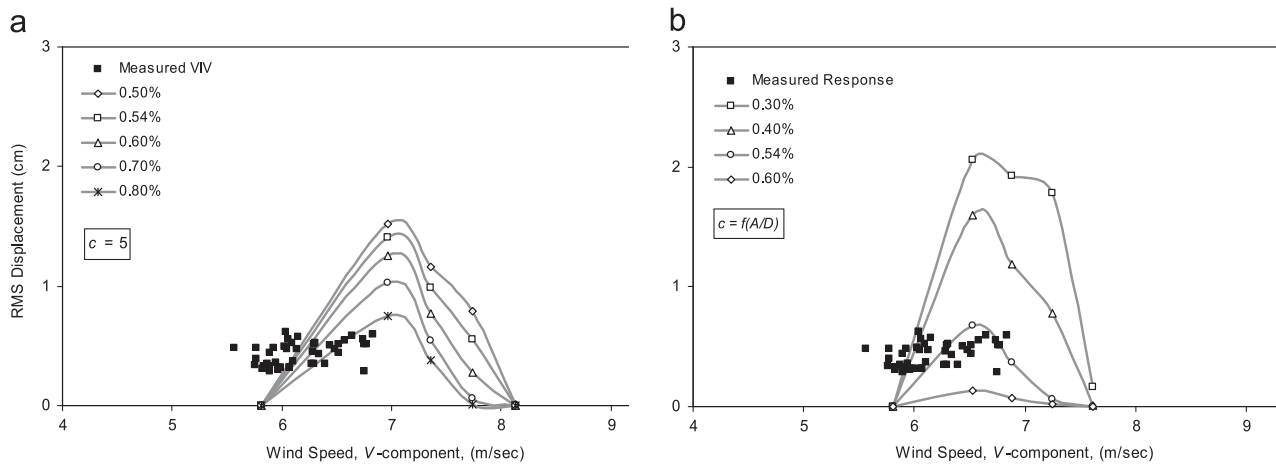


Fig. 12. Comparison between measured and simulated response in second vertical mode in region-II. (a) $c=5$, and (b) c function of A/D .

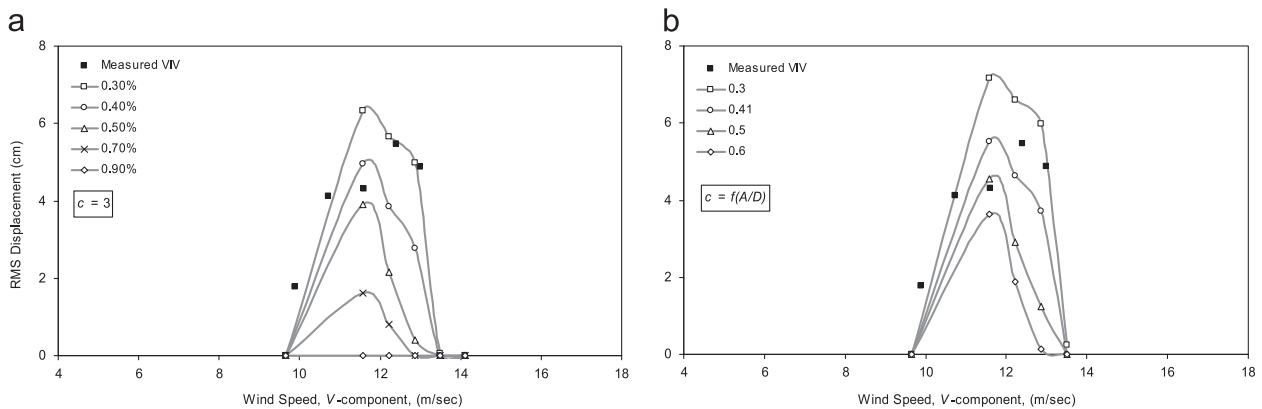


Fig. 13. Comparison among measured and simulated response in third vertical mode in region-III. (a) $c=3$. (b) c function of A/D .

maximum amplitude based on two observations. First, with increasing wind speeds, larger locked-in responses were observed on the bridge (Fig. 9). In other words, the maximum response in Region-III was higher than that of Region-I and the maximum response of Region-IV was higher than that of Region-III. In contrast, the responses in Region-II were found to deviate from this trend and the maximum response was smaller than that of Region-I. Second, this event of locked-in vibration occurred intermittently over a period of about 40 min only on December 11 and

12, 1999. Based on the observation of a single event, it cannot be stated with confidence that the bridge reached maximum response amplitudes in Region-II. The response in Region-II was simulated using the amplitude-dependent correlation model for several damping ratios. The coefficients of the correlation function were determined as $a=0.00564$ and $b=0.0135$.

The simulated response in the third vertical mode at midspan is presented in Fig. 13a. This locked-in oscillation corresponds to Region-III in Fig. 9a. The damping ratio in the third vertical mode

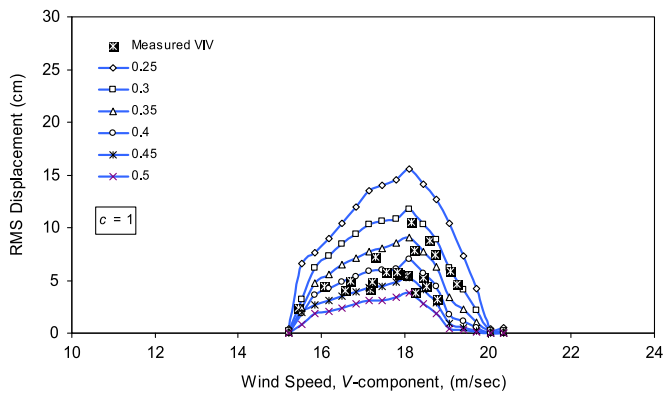


Fig. 14. Comparison among measured and simulated responses using aeroelastic coefficients of Set-II. Response of region-IV in first vertical mode with $c=1$.

was 0.41% and the maximum RMS response was 5.5 cm on the full-scale bridge. Among the various combinations of damping ratios and values of c , 0.41% damping with $c=3$ demonstrated good agreement with the measured response. The coefficients of the amplitude-dependent correlation function were determined as $a=0.00158$ and $b=0.0675$ (Fig. 13b).

5.3.2. Numerical response obtained from set-II aeroelastic coefficients

The measured midspan displacements of the bridge are also compared with the simulated responses obtained from aeroelastic coefficients of Set-II. A damping ratio of 0.3% and $c=1$ gave RMS response amplitudes that closely matched the full-scale responses of Region-IV (Fig. 14). However, the RMS response for 0.7% damping (which is the median damping ratio in first vertical mode) was calculated 0.5 cm, which was significantly smaller than the measured RMS response of 10.5 cm. These results can be explained by the fact that the aeroelastic coefficients of the leeward deck used in Set-II corresponded to non locked-in response and therefore had positive, constant damping properties at all amplitudes, which led to significantly lower aeroelastic forces on the leeward deck.

5.3.3. Response considering fully correlated aeroelastic forces

Using the median damping ratios measured on the bridge and aeroelastic coefficients of Set-I, the maximum response-amplitudes of the bridge were predicted as 4.7 cm in first vertical mode for Region-I, 5.4 cm in second vertical mode for region-II, 6.8 cm in third vertical mode for Region-III, and 15.5 cm in first vertical mode for Region-IV. These response amplitudes were higher than the maximum full-scale responses and can be considered an upper bound of the response as the assumption of full correlation predicts higher aeroelastic forces on a structure. However, the aeroelastic coefficients of Set-II yielded responses that were smaller than the measured responses.

6. Conclusions

The main goal of the research reported herein was to develop an analytical model capable of predicting the locked-in vibration of long-span bridges, an effort that has been proven to be challenging even for the most basic section such as a circular cylinder. The performance of the equation of motion critically depends on the reliable estimation of the five parameters of the equation. These are the aeroelastic coefficients, modal frequencies, mode shapes, modal damping ratio and correlation functions. An aeroelastic damping coefficient H_1^* and an aeroelastic stiffness coefficient H_4^* are required for the equation of motion. Formula to

identify these coefficients from wind tunnels tests of spring-mounted rigid models are presented and the aeroelastic coefficients of six bluff sections, three of which represent real bridge cross sections, are provided. The results show that H_1^* is a non-linear function of normalized amplitude. The shapes of H_1^* versus A/D curves vary with wind speeds as well as cross section profile. The H_4^* coefficients, on the other hand, do not depend on A/D . Both H_1^* and H_4^* depend on reduced frequency, K . H_1^* strongly governs the response of an oscillator during locked-in vibration. The complexity associated with correctly modeling a full-scale bridge for wind tunnel tests can affect the aeroelastic coefficients including the accuracy of the estimated lock-in regions. As the locked-in aeroelastic force on a real structure is not fully correlated along the span, response of the Fred Hartman bridge was simulated incorporating two different correlation models.

The extraction of the aeroelastic coefficients of the Fred Hartman bridge was particularly challenging due to the twin-deck configuration of the bridge. The aeroelastic coefficients were obtained from a simplified test setup, which captured the effect of the presence of a fixed section in the wake or upstream region of the oscillating section. The modal frequencies and modal damping ratios of the prototype bridge were obtained from full-scale measurements through rigorous analysis. A single-mode analysis was considered for numerical simulation supported by observations made in past full-scale measurements and wind tunnel experiments; current full-scale studies further supported this observation.

The responses using the correlation model of Eq. (21) with values of c ranging from 1 to 3, the aeroelastic coefficients of Set-I, and the estimated median modal damping ratios demonstrated good agreement with the full-scale results for all lock-in regions. If a test procedure similar to the one adopted in this research is used, it is recommended that the aeroelastic coefficients of Set-I be used for numerical simulation. The coefficients of the correlation model of Eq. (22) were obtained by fitting simulated response to full-scale response of the Fred Hartman bridge. These correlation coefficients can be used to simulate the response of other bridges with a section profile similar to that of the Fred Hartman bridge. It was observed in this study that the vortex-induced response depended strongly on structural damping ratio and correlation function; these two parameters remain significant sources of uncertainty in response predictions.

Unlike the previous analytical models that mainly attempted to match the response-amplitudes of a section model without giving much attention to the fluid-dynamics of the locked-in vibration, the model presented herein is based on sound physical arguments and has incorporated the important characteristics of locked-in vibration observed in wind tunnel experiments and in the field on full-scale structures. The assumptions and simplifications considered in the theoretical basis of the model are less likely to affect the quality of the predicted responses than the uncertainties associated with many of the parameters needed for the mathematical model.

References

- Billah, K.Y.R., Scanlan, R.H., 1991. Resonance, Tacoma Narrows Bridge failure, and undergraduate physics textbooks. *Am. J. Phys.* 59 (2), 118–124.
- Billah, K.Y.R., 1989. A Study of Vortex-Induced Vibration (Ph.D. dissertation). Princeton University, Princeton, NJ, USA.
- Bishop, R.E.D., Hassan, A.Y., 1964. The lift and drag forces on a circular cylinder in a flowing fluid. *Proc. R. Soc. Lond. Ser. A Math. Phys. Sci.* 277 (1368), 32–50.
- Blevins, R.D., 1990. Flow-induced Vibration, 2nd edition Van Nostrand Reinhold Publications, New York, NY, USA.
- Davenport, A.G., 1962. Buffeting of a suspension bridge by storm winds. *J. Struct. Div. ASCE* 88 (3), 233–252.
- Dowell, E.H., 1981. Non-linear oscillator models in bluff body aeroelasticity. *J. Sound Vib.* 75 (2), 215–264.

- Ehsan, F., Scanlan, R.H., 1990. Vortex-induced vibration of flexible bridges. *J. Eng. Mech. ASCE*, 116 (6), pp 1392–1411.
- Ehsan, F., 1989. The Vortex-Induced Response of Long, Suspended-Span Bridges (Ph.D. dissertation). the Johns Hopkins University, Baltimore, MD, USA.
- Feng, C.C., 1968. The Measurement of Vortex-Induced Effects in a Flow Past Stationary and Oscillating Circular and D-section Cylinders (M.Sc. thesis). University of British Columbia.
- Fujino, Y., Yoshida, Y., 2002. Wind induced vibration and control of the trans-Tokyo Bay Crossing Bridge. *J. Struct. Eng. ASCE* 128 (8), 1012–1025.
- Goswami, I., Scanlan, R.H., Jones, N.P., 1993. Vortex-Induced vibration of circular cylinders. II. New Model. *J. Eng. Mech. ASCE* 119 (11), 2288–2302.
- Hartlen, R.I., Currie, I.G., 1970. Lift oscillator model of vortex induced vibration. *J. Eng. Mech.* 96, 577–591.
- Irwin, P.A., 1998. The role of wind tunnel modeling in the prediction of wind effects on bridges. In: Larsen, A., Esdahl, S. (Ed.), *Proceedings of the International Symposium on Advances in Bridge Aerodynamics*, Denmark, pp. 99–117.
- Iwan, W.D., Blevins, R.D., 1974. A model for vortex-induced oscillations of structures. *J. Appl. Mech.* 41, 168–187.
- Jordan, D.W., Smith, P., 1987. *Nonlinear Ordinary Differential Equations*, 2nd edition Oxford University Press, New York.
- Landl, R., 1975. A mathematical model for vortex-induced vibration of bluff bodies. *J. Sound Vib.* 42 (20), 219–234.
- Larsen, A., 1993. Aerodynamic aspects of the final design of the 1624 m suspension bridge across the Great Belt. *J. Wind Eng. Ind. Aerodyn.* 48, 261–285.
- Larsen, A., Esdahl, S., Andersen, J.E., Vejrum, J.E.T., 2000. Storebaelt suspension bridge-vortex shedding excitation and mitigation by guide vanes. *J. Wind Eng. Ind. Aerodyn.* 88, 283–296.
- Macdonald, J.H.G., Irwin, P.A., Fletcher, M.S., 2002. Vortex-induced vibrations of the Second Severn Crossing cable-stayed bridge—full-scale and wind tunnel measurements. *Proc. Inst. Civ. Eng. Struct. Build.* 152 (2), 123–134.
- Main, J.A., Jones, N.P., 2001. Evaluation of viscous dampers for stay-cable vibration mitigation. *J. Bridge Eng.* 6 (6), 385–397.
- Mashnad, M., Jones, N.P., 2005. Experimental study on vortex-Induced vibrations of selected bluff sections, *Proceedings of the 10th Americas Conference on Wind Engineering*. Baton Rouge, USA.
- Mashnad, M., 2006. Modeling vortex-induced vibration of long-span bridges. University of Illinois at Urbana-Champaign, IL, USA.
- Matsumoto, M., Shirato, H., Araki, K., Haramura, T., Hashimoto, T., 2003. Spanwise coherence characteristics of surface pressure field on 2-D bluff bodies. *J. Wind Eng. Ind. Aerodyn.* 91, 155–163.
- Meirovitch, L., 1986. *Elements of Vibration Analysis*. McGraw-Hill, Inc..
- Ozkan, E., 2003. Evaluation of Response Prediction Methodology for Long-Span Bridges Using Full-Scale Measurements (Ph.D. dissertation). Johns Hopkins University.
- Sarpkaya, T., 1978. Vortex induced oscillation: a selective review. *J. Appl. Mech.* 46 (2), 241–256.
- Scanlan, R.H., 1998. Bridge flutter derivatives at vortex lock-in. *J. Struct. Eng. ASCE* 124 (4), 450–458.
- Scanlan, R.H., Jones, N.P., 1990. Aeroelastic analysis of cable-stayed bridges. *J. Struct. Eng.* 116 (2), 279–297.
- Scanlan, R.H., Tomko, J.J., 1971. Airfoil and bridge deck flutter derivatives. *J. Eng. Mech. Div. ASCE* 97 (EM), 1717–1737.
- Scanlan, R.H., 1981. State-of-the-art methods for calculating flutter, vortex-induced, and buffeting response of bridge structures. Federal Highway Administration, Report no. FHWA/RD-80/050.
- Simiu, E., Scanlan, R.H., 1986. *Wind Effects on Structures*. John Wiley, New York, USA.
- Skop, R.A., Griffin, R.A., 1975. On a theory for the vortex-induced oscillation of flexible cylindrical structures. *J. Sound Vib.* 41 (3), 263–274.
- Staubli, T., 1983. Calculation of the vibration of an elastically mounted cylinder using experimental data from a forced oscillation. *J. Fluid Eng. ASME* 105, 225–229.
- Van der Pol, B., 1926. On relaxation-oscillations. *Philos. Mag.* 7 (2), 978–992.
- Vickery, B.J., 1966. Fluctuating lift and drag on a long cylinder of square cross-section in a smooth and in a turbulent stream. *J. Fluid Mech.* 25, 481–494.
- Vickery, B.J., Basu, R.I., 1983. Across wind vibration of structures of circular cross section. Part I. Development of a mathematical model for two-dimensional conditions. *J. Wind Eng. Ind. Aerodyn.* 12, 49–73.